

Shoulder Flexion (Lat Length Test)

Screening Test	Description
 Shoulder flexion (lat length test)	Client position: • Standing against a wall or lying supine on the floor, or table, with hips and legs straight • If more comfortable, hips flexed, knees bent, feet flat, palms facing each other Type of motion: • Passive (lying) • Active (standing) Target motion: • Client brings both arms above the head (flexion), as far as possible, then passively relaxes in that position Verbal instructions: • “Raise both arms above your head, as far as possible, then relax while holding that position.” Assessment: • Normal mobility: Upper arms extend directly overhead in line with the lateral midline of the torso without impairment • Restricted shoulder flexion: Client is unable to reach their arms in line with their torso, or displays impairments such as: ○ Elbow flexion ○ Lumbar extension ○ Rib cage flaring ○ Shoulder elevation (shrugging) ▪ Overactive/shortened: Latissimus dorsi, teres major, and pectoralis major (lower fibers)

Shoulder Retraction (Pectoralis Minor Test)

Screening Test	Description
 Shoulder retraction (pectoralis minor test)	Client position: Lying supine on a table with legs extended Type of motion: Passive Target motion: Client rests both arms at the side of the body with elbows extended and palms facing upward Verbal instructions: - “Relax both arms at the side of your body with your elbows extended and palms up.” Assessment: • Normal mobility: Client can rest their shoulders flat on the table without impairment in other kinetic chain checkpoints • Restricted shoulder retraction: Client’s shoulders will rest at a level off the table or display an impairment such as: ○ Lumbar spine extension ○ Cervical extension ○ Rib cage flaring ○ Accessory motion of the arms ▪ Overactive/shortened: Pectoralis minor on the same side as the elevated shoulder or impairment

Shoulder and Thoracic Mobility Tests

Shoulder Extension

Screening Test	Description
	Client position: • Standing • Lying supine on a table Type of motion: • Passive (lying) • Active (standing) Target motion: • Client reaches the arm rearward, as far as is comfortable, while keeping the elbow and wrist straight Verbal instructions: • “Reach your arm backward as far as you can, then repeat with the other arm.” Assessment: • Normal mobility: Client is able to reach just beyond roughly 45 degrees of extension (halfway between the line of the horizon and the wall) without impairment • Restricted flexion: Client is unable to reach the targeted mobility benchmark, or displays impairments such as elbow flexion, shoulder elevation, or the scapula of the test shoulder tilting upward and protracting ○ Overactive/shortened: Anterior deltoid, pectoralis major (upper fibers), coracobrachialis, biceps brachii

Shoulder extension

Shoulder Internal and External Rotation

Screening Test	Description
	Client position: • Standing or lying supine on the floor, or a table, with hips and legs straight • If they are more comfortable with hips flexed, knees bent and feet flat Type of motion: Active Target motion: • With shoulders and elbows abducted, and bent to 90 degrees, client rotates arms downward to the table or floor; rotation occurs only at the glenohumeral joint • Then, client rotates arms backward, attempting to touch the back of their hand to the plane of the table or to the floor Verbal instructions: • “With your arms out to the side and elbows bent, rotate your arms and hands forward toward the table. Then, rotate your arms and hands backward toward the table.” Assessment: • Normal mobility: ○ Internal rotation: Client can rotate the forearm/hand forward to near the table (roughly halfway between 45 degrees and the table) without impairment ○ External rotation: Client can rotate the forearm/hand backward to nearly touching or touching the plane of the table without impairment • Restricted internal rotation: Client is unable to reach the targeted mobility benchmark, or displays impairments such as the test shoulder elevating and protracting off the table, a change in elbow flexion, or wrist flexion ○ Overactive/shortened: Teres minor and infraspinatus • Restricted external rotation: Client is unable to reach the target mobility benchmark, or displays impairments such as a change in elbow flexion, shoulder elevation, rib cage flaring, or wrist extension ○ Overactive/shorted: Subscapularis, teres major, latissimus dorsi, and pectoralis major

Shoulder internal/external rotation

Shoulder and Thoracic Mobility Tests

Thoracic Extension

Screening Test	Description
	Client position: • Seated on a standard, low-backed chair with hands and arms crossed over chest • Lumbar and cervical spines neutral throughout • Chair back should end just beneath the client's shoulder blades Type of motion: Active Target motion: Client arches their mid-back over the chair as far as comfortably possible Verbal instructions: • "Cross your arms over your chest. While keeping your ribs and low back in place, and your neck in line with your torso, lean your mid-back over the chair as far as you can without tipping the chair." Assessment: • Normal mobility: The upper back and head tilt backward, where the tops of the shoulders extend past the chair back (roughly 25 degrees of thoracic extension), and the sternum is nearly parallel to the horizon without impairment • Restricted thoracic extension: Client is unable to reach the targeted mobility benchmark, or displays impairments such as: ○ Cervical extension ○ Ribs flaring ○ Lumbar extension ▪ Overactive/shortened: Rectus abdominis, internal/external obliques

Thoracic extension

Seated Thoracic Rotation

Screening Test	Description
	Client position: • Seated with hands crossed in front of body, or crossed with holding a stick or dowel rod • Place a medicine ball or foam roller between the knees to stabilize the lower body • Ensure the shoulder blades are retracted and depressed Type of motion: Active Target motion: • Client squeezes the ball or roller between the knees to lock the hips in place • Client maintains a neutral cervical spine • Client then rotates the upper body to each side, as far as possible Verbal instructions: • “Hold the stick to your chest under your crossed arms. Squeeze the roller between your knees, look forward, and keep your nose in line with your sternum. Rotate your trunk to the right as far as you can. Then, rotate to the left.” Assessment: • Normal mobility: The sternum (or stick) rotates roughly 45 degrees from the starting position to each side without impairment • Restricted thoracic rotation: Client is unable to reach the targeted mobility benchmark, or displays impairments such as: ○ Lateral flexion of the spine ○ Leaning forward or backward ○ Shoulder protraction ▪ Overactive/shortened: Rectus abdominis, internal/external obliques, and erector spinae on the side opposite of the restriction

Seated thoracic rotation